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CINEC ROBOTICS CHALLENGE '23

FACULTY OF ENGINEERING AND TECHNOLOGY

CONTENT

Section 01:

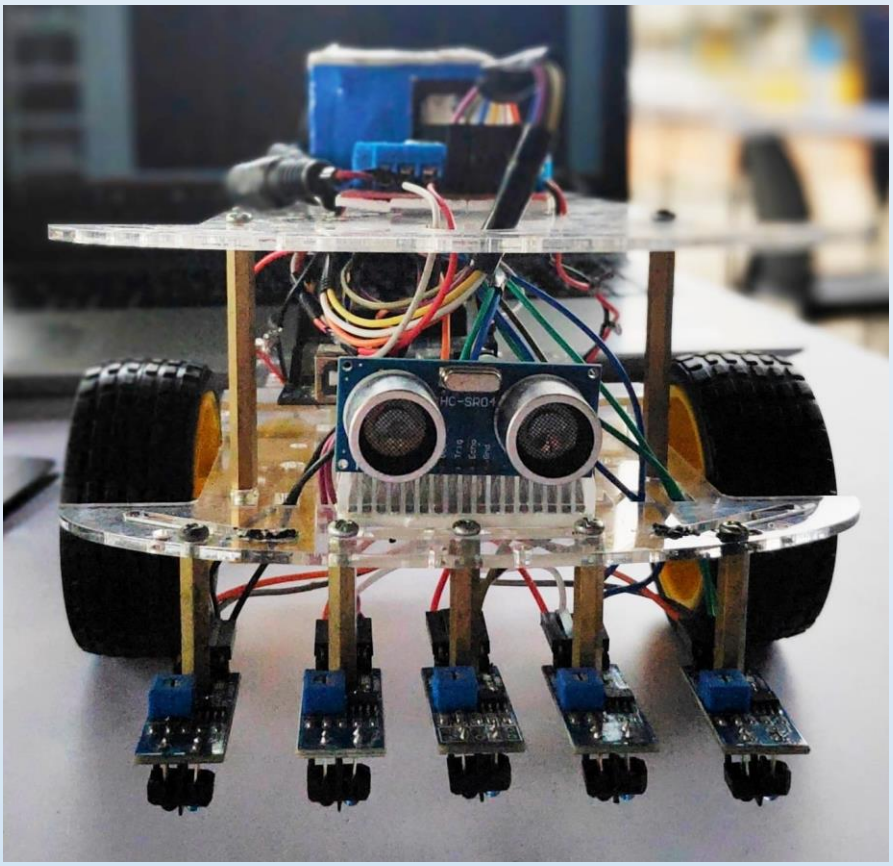
LINE FOLLOWING ROBOT (Age Category – Under 12)

Section 02:

MAZE SOLVING ROBOT (Age Category – Between 12 - 20)

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THE WARRIORBOTS



LINE FOLLOWING ROBOT

(Age Category – Under 12)

INTRODUCTION

Robotics and automation is one of the fastest emerging fields all over the globe and both international and local communities are enthusiastically focusing on the innovative technologies related to robotics. The Engineering faculty of CINEC campus has launched the "WARRIOR BOTS" robotics challenge for the first time to provide a platform for the talented school students to learn and present their passion and skills in the field of robotics.

Line following robot is targeted for the school students in the age category below 12 years old. Your task is to design and build an autonomous robot that can successfully conquer the given challenge by strictly adhering to the guidelines below.

Based on the design and performance of the robots, an external panel of judges will adjudicate the competition. The decision of the judges will be the final and irrevocable decision.

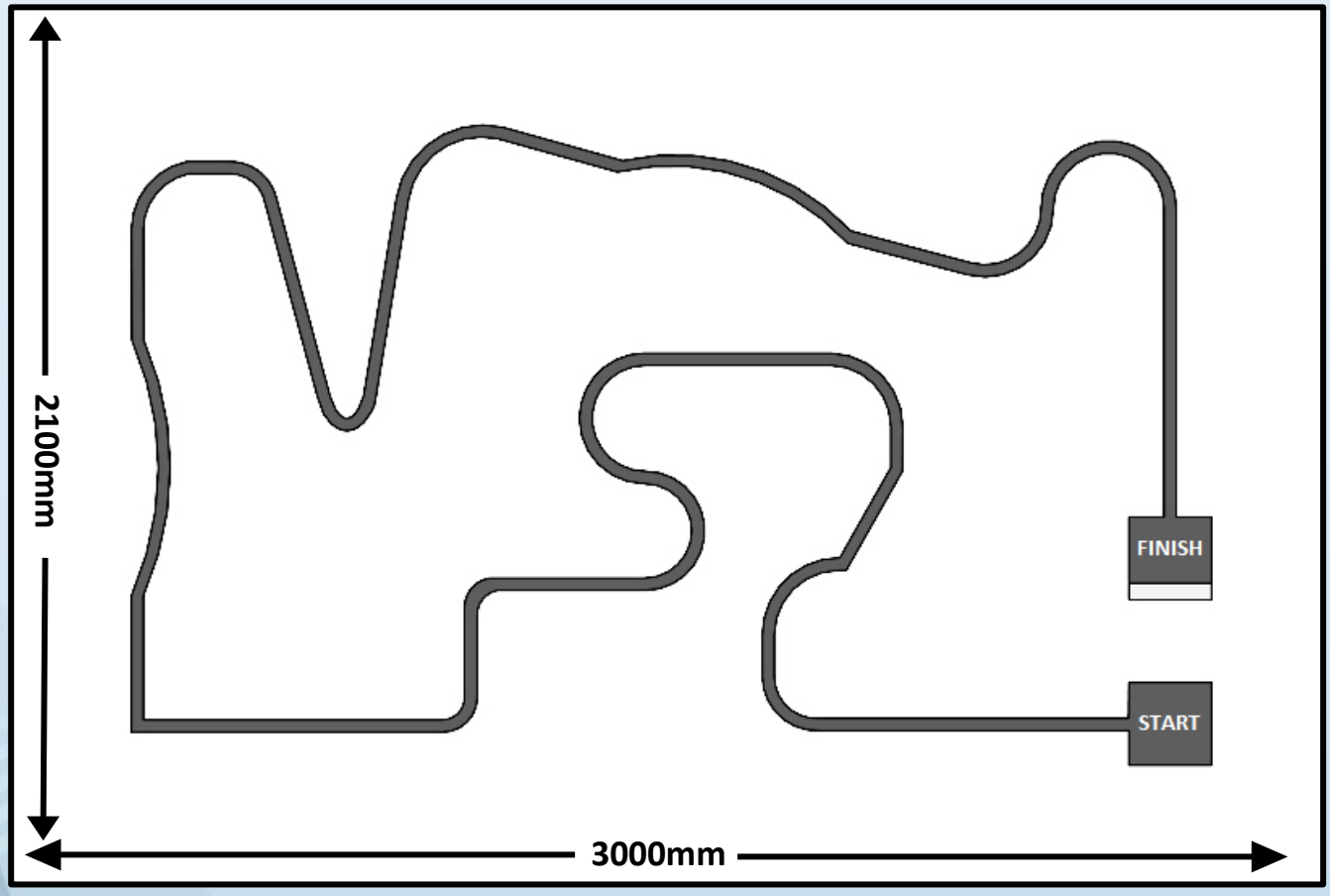
Moreover, please note that any amendments to the rules or the task will be updated via e-mail. If you have any questions or clarifications, you may contact the "WARRIOR BOTS" organizers through email warriorbotscomp@gmail.com or phone(s) 0770348841 (Pramod Palayangoda) / 0710640499 (Yasindu Karunaratne).

1.0 TASK - Line Following

At the beginning of the task, the robot will be placed on the starting square. First, the robot has to follow a black line on a white surface. These path may contain straight lines, curved lines and 90° degree corners. All the line segments you encounter throughout the task will be of similar specifications as described later.

2.0 ARENA SPECIFICATIONS

2.1 Full view



2.2 Key Arena Specifications

- The sample view of the arena is shown in the following figure 01.
- Start and Finish squares are black squares with the parameters of 20cm × 20cm and at the bottom of the Finish square there will be a 4cm white line to stop the robot.
- The floor will be covered fully with non-glossy white colour sticker. The track will be black colour line with a width of 3cm.
- The minimum gap between the boundaries and the track is 20cm. All the dimensions may be subjected to the maximum error margin of 3mm.
- There are following main 4 sections where the robot should complete and finish the run.

2.3 Line Segments



90° Turn – The maximum distance between edges of the conner will be 4.3cm.

Half Circle Turn – The maximum distance between the inner diameter will be 26.8cm.





V – Turn - The effective distance between the inner starting points will be 8cm.

S – Turn – The junction consisted with two half circles, each with the diameter of 28.65cm.



3.0 ROBOT SPECIFICATIONS

- Dimensions of the robot should not exceed 20cm × 20cm (width × length). The robot must be inside of the start box before the race starts.
- The robot should be completely autonomous. Using manual controlling methods will lead to disqualification of the robot.
- The robot should be powered with an internal power source not exceeding 15V.
- The robot should be built entirely by the team members. Therefore, outsourced, pre-made or Lego robot design will not be allowed except readymade microcontrollers, sensors, motors or other electronic devices.
- The robot should not cause any damage to the arena and any damage to the arena leads to disqualification. If the judges feel the robot has a high risk of damaging the arena, they have authorization to stop the run.
- The robot should have a simple starting procedure such as a single switch to power up the robot. There cannot be external force or pulse involving on robot in the starting procedures.
- The robot should operate under provided lighting conditions. The robot cannot be transformed into different parts.

4.0 RULES AND REGULATIONS

4.1 General

- There won't be any arena changes once the round has started. All team will have the same arena.
- All the teams must submit their robots to the organizers 15 minutes before the start of the first round. After that, the robot will be given to the relevant participant only for its attempts at the round.
- The contestants must be prepared to start within 5 minutes after the call; it not, the attempt is lost.
- A team should place the robot entirely inside the starting square at the start of their run. When the judges give the signal, the robot can be switched on. From then on, the robot should navigate autonomously. The contestants should not manually alter the orientation of the robot during the gameplay. Additionally, the contestants should not communicate with or control the robot during an attempt.
- A maximum of 3 attempts are given in a single round, and the overall time (the total time of all attempts taken) will be counted.
- A maximum time of 15 minutes (period of gameplay) is allocated per team. Therefore, all the attempts are reserved for this period only. If the robot exceeds the time limit of 15 minutes, your robot will be removed from the arena.
- Program or hardware modifications of the robot are not allowed within this 15 minutes.
- The time taken to travel from the start square to the End square is called the total run time. The clock will start when the judges give the signal to start. The clock will stop when the robot reaches the End square, and a proper indication is issued by the robot that it has completed the task
- The clock will not be paused during an attempt.
- If the robot drifts out of the line to the extent that no part is on top of the line in line following segments, the judges will consider it as jumping out of line. However, if the robot finds its way back to the line on itself, it can continue, provided that the distance skipped by the robot along the line is less than 30 cm. The judges may deduct points in this case. If the robot does not find its way back to the line within a skipped distance of 30 cm. which would be considered the end of that attempt, you will be allowed to remove the robot from the arena.

4.2 Calibration

- Two additional minutes before the gameplay period of 15 minutes is given for the calibration of the robot.
- The robot can only use the arena from the starting square until the end of line segment 1 for calibration.
- Calibrations can only be done through external adjustments of the robot. Therefore, program changes nor hardware part replacements are not allowed.
- The team can request the start of their first attempt before the calibration period ends. The gameplay period of 15 minutes will start at that moment. (The remaining calibration time will not be added to the 15 minutes given for the three attempts)
- If a team fails to finish calibrating within these 2 minutes, the extra time taken will be deducted from the period of gameplay of 15 minutes.

5.0 TEAM COMPOSITION AND ELIGIBILITY

- A team may be comprised of maximum up to five (05) members. Students from different schools can get together to form a team but the team registration will be done only under one school name.
- The members of the team should be less than 12 years of age.
- All team members should be school (local/international) students.
- All team members should produce a document of identity (birth certificate or National Identity Card) to prove the eligibility to participate in the competition.
- One team can only submit one robot for the competition.
- Violation of any of the above conditions may lead to disqualification.

6.0 JUDGING

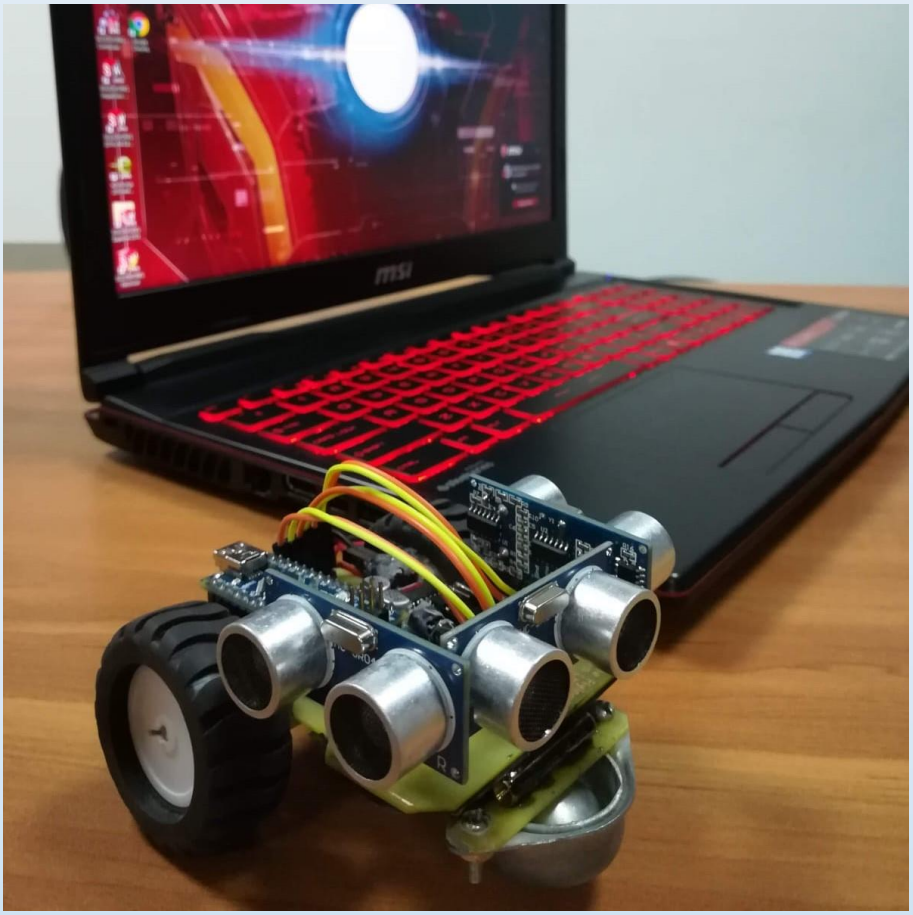
- The winning team who successfully completes the assigned task will be selected based on a well-defined marking rubric that will be followed by the panel of judges.
- The judges may decide to assign an extra task which may help them to arrive at a final decision (not mandatory) and this task will not be pre-defined.
- The robot's coding may be checked upon judges' request.
- The team members will be questioned about the robot design and all team members must be able to answer the questions.
- Penalties will be given if the robot deviates less than 30cm in length without covering any part of the line following arena.
- Final judging criteria will be made available later.
- The team will be given three (03) attempts and if the judges are not satisfied with the performance of the robot, they may ask to stop that particular attempt.
- The decision of the panel of the judges will be considered as the final and irrevocable decision.

PLEASE REFER THE FOLLOWING LINK TO THE REGISTRATE FOR THE COMPETITION

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CINEC CAMPUS
Beyond A Graduate

THE WARRIORBOTS



MAZE SOLVING ROBOT

(Age Category – Between12 to 20)

INTRODUCTION

Robotics and automation is one of the fastest emerging fields all over the globe and both international and local communities are enthusiastically focusing on the innovative technologies related to robotics. The Engineering faculty of CINEC campus has launched the "WARRIOR BOTS" robotics challenge for the first time to provide a platform for the talented school students to learn and present their passion and skills in the field of robotics.

Maze solving robot is targeted for the school students in the age category between 12-20 years old. Your task is to design and build an autonomous robot that can successfully conquer the given challenge by strictly adhering to the below mentioned guidelines. The challenge mainly consists of a maze solving main task and a colour detection subtask.

Based on the design and performance of the robots, an external panel of judges will adjudicate the competition. The decision of the judges will be the final and irrevocable decision.

Moreover, please note that any amendments to the rules or the task will be updated via e-mail. If you have any questions or clarifications, you may contact the "WARRIOR BOTS" organizers through email warriorbotscomp@gmail.com or phone(s) 0770348841 (Pramod Palayangoda) / 0710640499 (Yasindu Karunaratne).

1. TASK – Solving Maze

The objective for the contestants is to build a robot that is capable of exploring different maze configurations and to work out the most efficient route with the shortest travel time from start to finish. The subtask is to identify the final destination point using colour detection technology.

2. GENERAL RULES

- The competitors must be prepared to start within 5 minutes after the call and if not, the run is lost.

- The robot has 7 minutes to complete the task. During this time, it can try to find more than one path. Maximum number of runs are limited to 5.

- The time taken to travel from the start square to the destination square is called the 'run' time. Travelling from the destination square back to the start square is not considered as a run. The total time taken from the first activation of the maze solving robot until the completion of each run is limited to 7 minutes. If the robot touches the maze walls any time during the contest, they are considered as 'touched' and it's 'run' time is penalized with 2 seconds. Scoring is based on these parameters. Touching the robot during a run by an operator is not allowed and will lead to terminating the present run.

- The total time on the maze (maze or search time), which is 7 minutes shall be measured from the time the robot is first activated. The time for each run (run time) shall be measured from the moment the robot leaves the start square until it enters the destination square. The run time will start when front edge of the robot crosses the start square and stops when the robot detected the red square at the final destination point.

- Every time the robot leaves the start square, a new run begins. If the robot has not entered the destination square, the previous run is aborted. For example, if a robot re-enters the start square (before entering the destination square) on a run, that run is aborted, and a new run will be deemed begun, with a new time that starts when the starting square is exited.

- If the robot reaches the destination square, it may stop and remain at the maze centre, or it may continue to explore other parts of the maze or make its own way back to the start. When the mouse chooses to stop at the final destination point, it may be lifted out, manually, and restarted by the handler and it will be considered as a restart penalty.

- Competitors can carry out either a "Search Run" or a " Fast Run" where in search run robot can trace all the paths from the starting square to the destination square followed by a speed run in which robot moves through the optimum path determined by the search run.

- After the search run a team is allowed to take up to 5 attempts to reach the destination square.

- Each robot is allowed a maximum of 7 minutes to perform in the platform. The judges have the discretion to request a robot to retire early if by its lack of progress, it has become boring, or if by erratic behavior it is endangering the state of the maze.

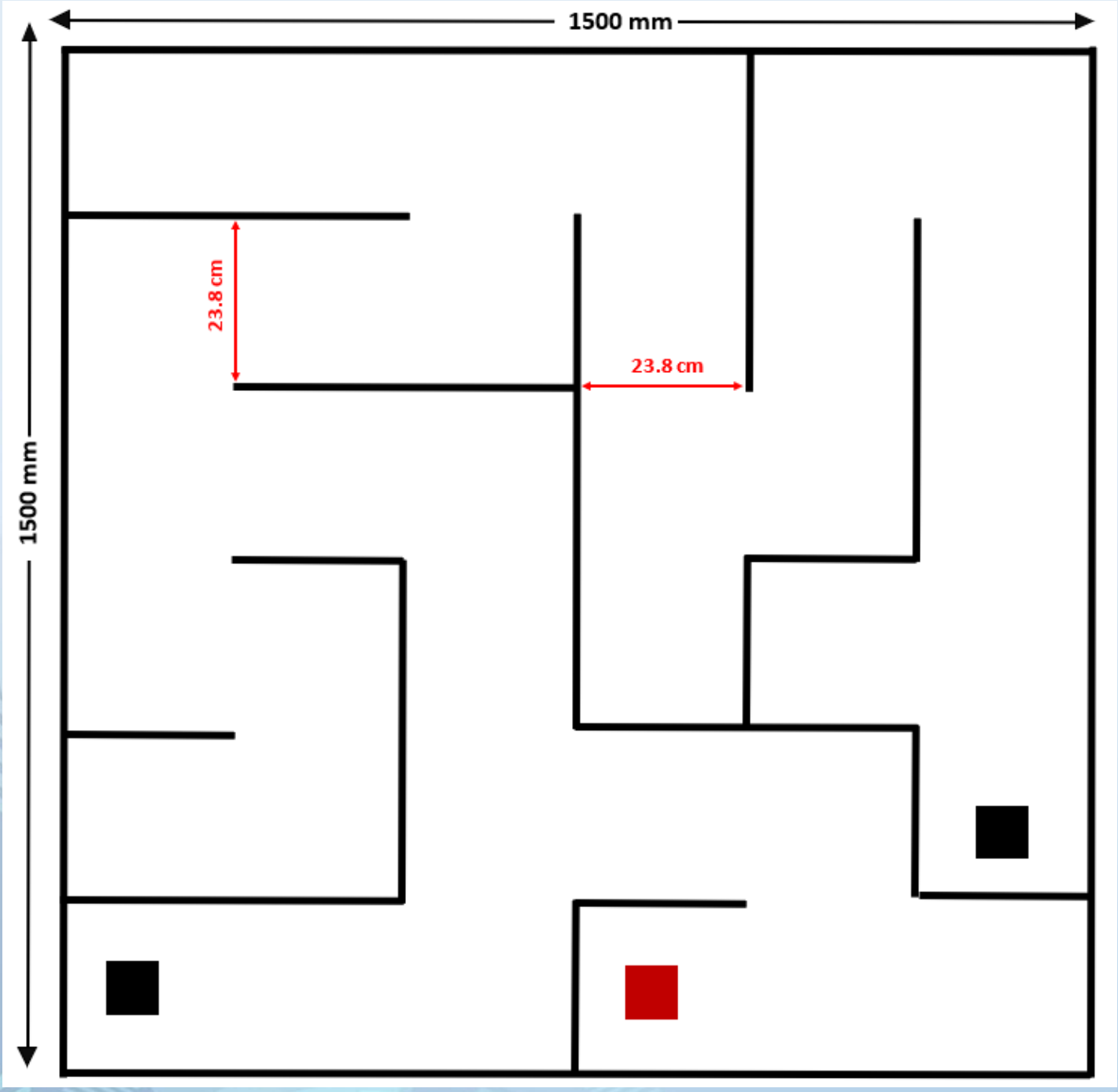


Figure 1: Maze Arena

3. PLAYING FIELD (MAZE)

- The maze shall comprise 6 x 6 multiples of a 25 cm x 25 cm unit square as shown in Figure 1. The walls constituting the maze shall be 7 cm high and 1.2 cm thick. Passageways between the walls shall be 23.8 cm wide. The outside wall shall enclose the entire maze.

- The side of the maze walls shall be white. The floor of the maze shall be made of wood and finished with a non-glossy white sticker.

  WARNING: Do not assume the walls are consistently white, or that the floor is consistently white. Fading may occur; parts from different mazes may be used. Do not assume the floor provides a given amount of friction. It is simply stickered Melamine board and may be quite slick. The maze floor may be constructed using multiple sheets of Melamine. Therefore, there may be a seam between the two sheets on which any low-hanging parts of a robot may snag.

- The start square of the maze shall be located at one of the corners. There are two black squares and one red square in the maze floor and the destination point is the red square and robot need to have the ability to identify that red square as the final destination.

- Square posts, each 25cm x 1.2cm x 7cm high, at the four corners of each unit square are called lattice points. The maze shall be constituted such that there is at least one wall touching each lattice point, except for the destination square.

- The dimensions of the maze shall be accurate to within 5% or 2cm, whichever is less. Assembly joints on the maze floor shall not involve steps of greater than 0.5mm. The change of slope at an assembly joint shall not be greater than 4 degrees. Gaps between the walls of adjacent squares shall not be greater than 2 mm.

- Multiple paths to the destination square are allowed and are to be expected.

- Lighting conditions of the stage may vary and competitors should adjust their robots accordingly.

- Only 2 persons from a team can involve in adjusting and setting the robot on the arena and other team members can also stay on the stage without getting involved in the arena.

- You will be given a sample maze to carry out any test runs or any calibrations from a week before the competition. Time slots for the test runs will be allocated and informed later.

- If the robot is reached to the final destination (Red square) and not able to reach back to start position, one team member can position robot at the start point and it will lead to a restart penalty.

4. ROBOT'S SPECIFCATIONS

- Maze solving robot must be completely self-contained and must receive no outside assistance.
- The method of propulsion is at the discretion of the builder, provided that the power source is non-polluting - internal combustion engines would probably be disqualified on this count.
- Although the superstructure of the robot may 'bulge' above the top of the maze walls, robot must be subjected to the following size constraints - width 20 cm, length 20 cm. There is no height limit.
- The method of wall sensing is at the discretion of the builder; however, the robot must not exert a force on any wall which is likely to cause damage.
- If the judges, consider that a robot has a high risk of damaging or sullying the maze they will not permit it to run. Nothing may be deposited in the maze. The robot must negotiate the maze; it must not jump over, climb, scratch, damage or destroy the walls of the maze.

5. INITIAL QUALIFICATION

- Size within the specified limit (Wooden square should be move without touching the robot)
- No RF modules.
- Battery voltage should be less than 15V.
- Robot shouldn't damage the arena and judges have the full authority to terminate the run in such cases.
- No loosing parts.
- No leaking oils or any substances.
- Manual switches are not allowed to change the status of navigation algorithm.

6. TEAM COMPOSITION AND ELIGIBILITY

- A team may be comprised of maximum up to five (05) members. Students from different schools can get together to form a team but the team registration will be done only under one school name.
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